



Networks Neural Physics-Informed

Problem Instability Stream Two Solving For Framework A

Shetaie Hosein

۱۴۰۴ Aban



تست فونت فارسی

°

- ◀ این یک اسلاید آزمایشی برای بررسی نمایش درست متن فارسی است
- ◀ اعداد فارسی: ۰۱۲۳۴۵۶۷۸۹
- ◀ ترکیب فارسی و لاتین: Vazirmatn و xepersian



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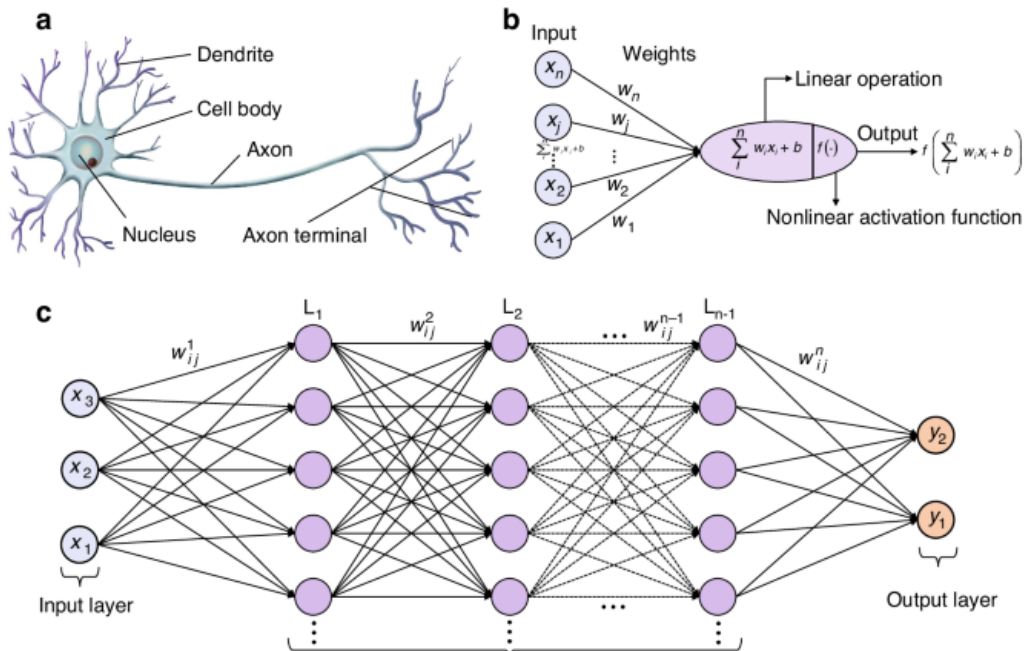
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Networks Neural

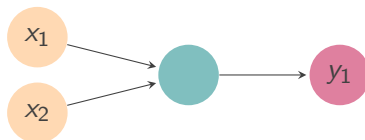
NN to Introduction 1





logic AND for Networks

NN to Introduction 1



$$y_1 = f(w_1x_1 + w_2x_2 + b)$$

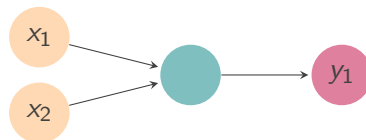
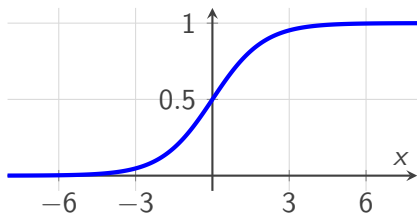
$y_1 = x_1 \wedge x_2$	x_2	x_1
0	0	0
0	1	0
0	0	1
1	1	1



logic AND for Networks

NN to Introduction 1

$$f(x) = \frac{1}{1 + e^{-x}}$$



$$y_1 = f(w_1x_1 + w_2x_2 + b)$$

$y_1 = x_1 \wedge x_2$	x_2	x_1
0	0	0
0	1	0
0	0	1
1	1	1



logic AND for Networks

NN to Introduction 1

$$y_1 = f(5.5x_1 + 5.5x_2 - 8)$$

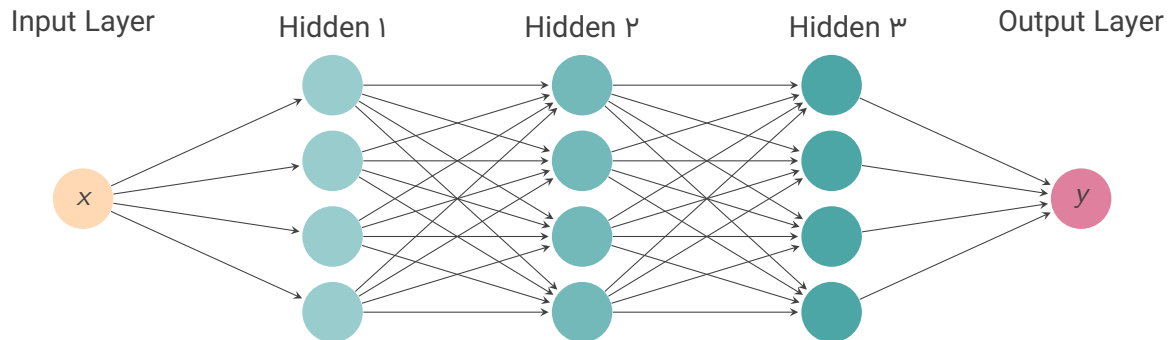
Output	$f(\text{sum})$	Sum Weighted	x_2	x_1
0	0003.0	0.8-	0	0
0	075.0	5.2-	1	0
0	075.0	5.2-	0	1
1	952.0	0.3	1	1

جدول: AND for output network Neural operation.



Networks Neural a Training

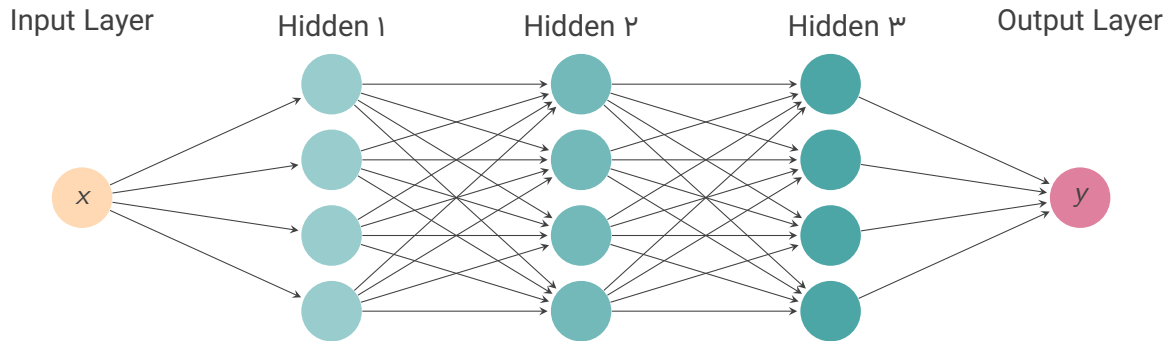
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Networks Neural a Training

NN to Introduction 1



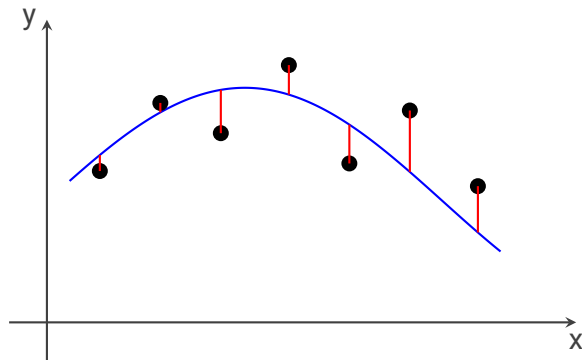
$$\text{Loss} = \frac{1}{N} \sum_i^N (y_{\text{nn}}(x_i) - y_{\text{true}}(x_i))^2$$



Networks Neural a Training

NN to Introduction 1

$$\text{Cost} = \frac{1}{N} \sum_i^N (y_{\text{nn}}(x_i) - y_{\text{true}}(x_i))^2$$

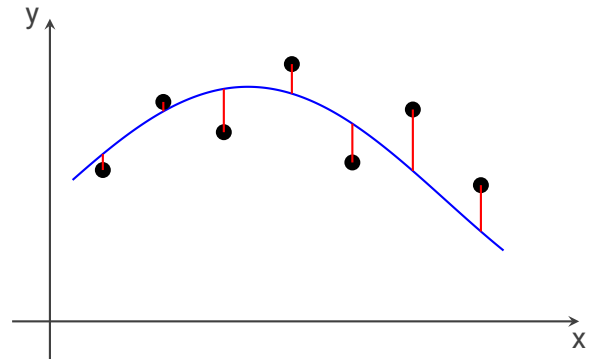
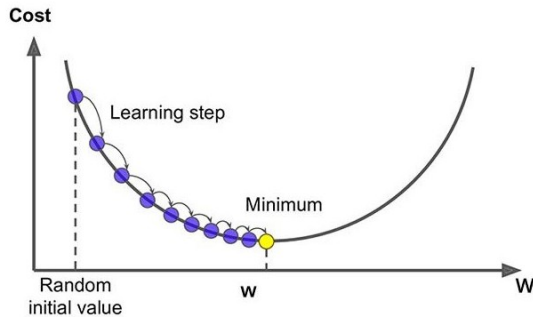




Networks Neural a Training

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$$\text{Cost} = \frac{1}{N} \sum_i^N (y_{\text{nn}}(x_i) - y_{\text{true}}(x_i))^2$$





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Learning Physics-Driven to Data-Driven From

Networks(PINNs) Neural Informed Physics ۲

Models Data-Driven Purely of Limitations The

datasets high-quality large, Require ◀

noisy or scarce are data when poorly Perform ◀

momentum) energy, mass, of conservation (e.g., laws physical violate May ◀

Learning Physics-Guided for Motivation

known are laws but limited are data problems, scientific In ◀

learning guide to constraints) (PDEs, knowledge domain Use ◀

function loss the into directly physics Integrate ◀

alone data of instead physics + data both from Learn ◀



PINNs

Networks(PINNs) Neural Informed Physics ۲



Journal of Computational Physics

Volume 378, 1 February 2019, Pages 686-707



Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations

M. Raissi ^a, P. Perdikaris ^b , G.E. Karniadakis ^a

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Highlights

- We put forth a deep learning framework that enables the synergistic combination of mathematical models and data.
- We introduce an effective mechanism for regularizing the training

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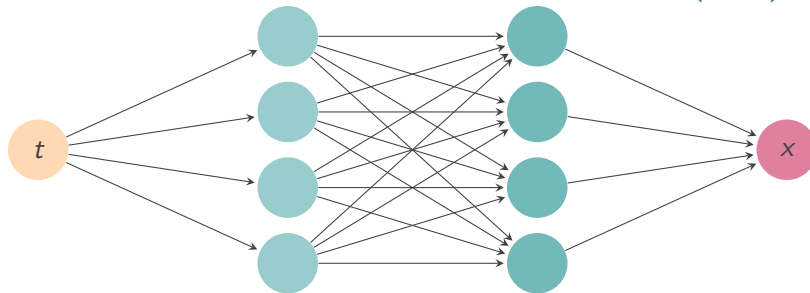


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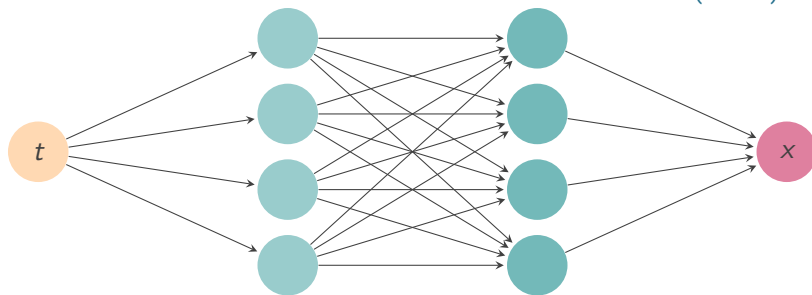


NN Naive

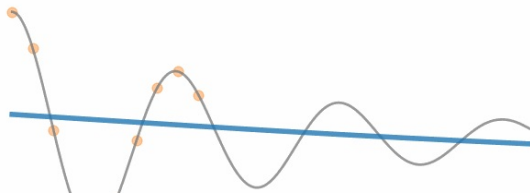
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$$\text{Loss} = \frac{1}{N} \sum_i^N (x_{\text{nn}}(t_i) - x_{\text{true}}(t_i))^2$$



$$\text{Loss} = \frac{1}{N} \sum_i^N (x_{\text{nn}}(t_i) - x_{\text{true}}(t_i))^2$$



Training step: 10

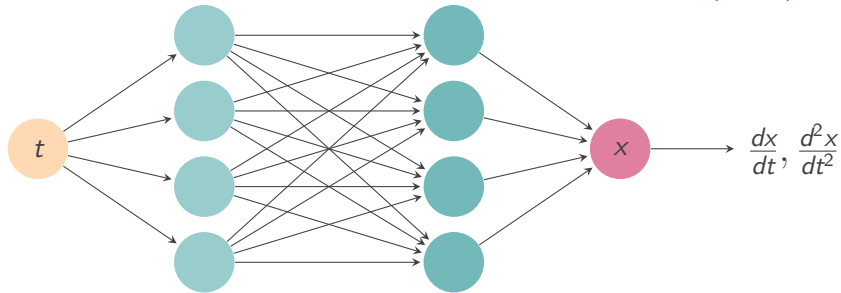
- Exact solution
- Neural network prediction
- Training data

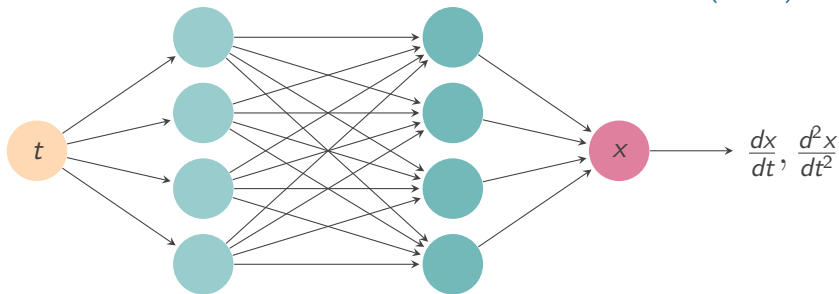


Oscillator

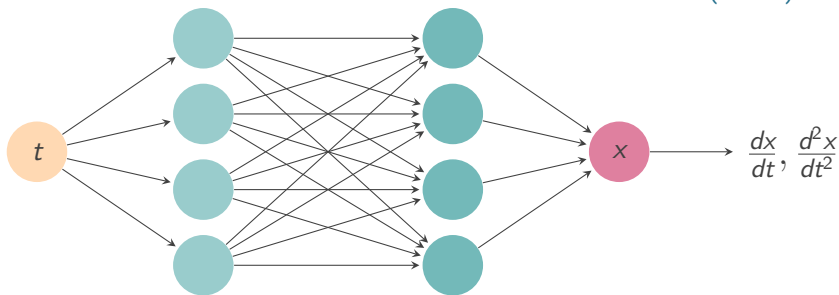
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$$m \frac{d^2 x}{dt^2} + \mu \frac{dx}{dt} + kx = 0$$

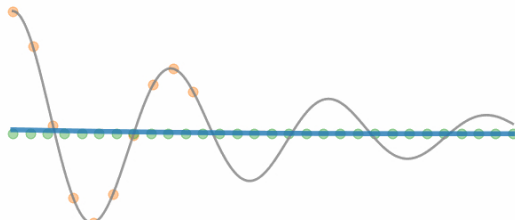




$$\text{Loss} = \frac{1}{N} \sum_i^N (x_{\text{nn}}(t_i) - x_{\text{true}}(t_i))^2 + \frac{1}{M} \sum_j^M \left(\left[m \frac{d^2}{dt^2} + \mu \frac{d}{dt} + k \right] x_{\text{nn}}(t_j) \right)^2$$



$$\text{Loss} = \frac{1}{N} \sum_i^N (x_{\text{nn}}(t_i) - x_{\text{true}}(t_i))^2 + \frac{1}{M} \sum_j^M \left(\left[m \frac{d^2}{dt^2} + \mu \frac{d}{dt} + k \right] x_{\text{nn}}(t_j) \right)^2$$



Training step: 150

- Exact solution
- Neural network prediction
- Training data
- Physics loss training locations



problem Forward

Networks(PINNs) Neural Informed Physics ۲

the and conditions boundary or and initial the only knowing by equation differential a Solving



problem Forward

Networks(PINNs) Neural Informed Physics ۲

the and conditions boundary or and initial the only knowing by equation differential a Solving
data. indirect or partial from them infer to is goal

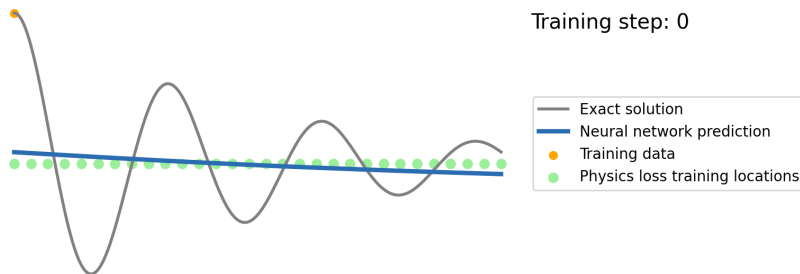
$$\text{Loss} = (x_{nn}(t=0) - 1)^2 + \lambda_1 \left(\frac{dx_{nn}}{dt}(t=0) \right)^2 + \frac{\lambda_2}{N} \sum_j^N \left(\left[m \frac{d^2}{dt^2} + \mu \frac{d}{dt} + k \right] x_{nn}(t_j) \right)^2$$

problem Forward

Networks(PINNs) Neural Informed Physics ۲

the and conditions boundary or and initial the only knowing by equation differential a Solving
data. indirect or partial from them infer to is goal

$$\text{Loss} = (x_{\text{nn}}(t=0) - 1)^2 + \lambda_1 \left(\frac{dx_{\text{nn}}}{dt}(t=0) \right)^2 + \frac{\lambda_2}{N} \sum_j^N \left(\left[m \frac{d^2}{dt^2} + \mu \frac{d}{dt} + k \right] x_{\text{nn}}(t_j) \right)^2$$





problem Inverse

Networks(PINNs) Neural Informed Physics ۲

the and unknown are it of parts some but known, is form equation the problem, inverse an In



problem Inverse

Networks(PINNs) Neural Informed Physics ۲

the and unknown are it of parts some but known, is form equation the problem, inverse an In
data. indirect or partial from them infer to is goal

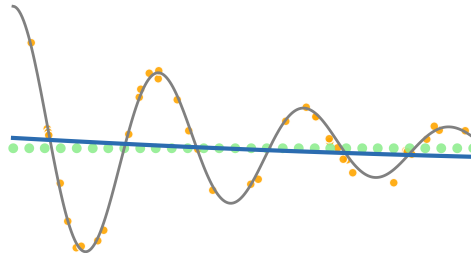
$$\text{Loss} = \frac{\lambda}{N} \sum_i^N (x_{\text{nn}}(t_i) - x_{\text{true}}(t_i))^2 + \frac{1}{M} \sum_j^M \left(\left[m \frac{d^2}{dt^2} + \mu \frac{d}{dt} + k \right] x_{\text{nn}}(t_j) \right)^2$$

problem Inverse

Networks(PINNs) Neural Informed Physics ۲

the and unknown are it of parts some but known, is form equation the problem, inverse an In
data. indirect or partial from them infer to is goal

$$\text{Loss} = \frac{\lambda}{N} \sum_i^N (x_{\text{nn}}(t_i) - x_{\text{true}}(t_i))^2 + \frac{1}{M} \sum_j^M \left(\left[m \frac{d^2}{dt^2} + \mu \frac{d}{dt} + k \right] x_{\text{nn}}(t_j) \right)^2$$



Training step: 0

- Exact solution
- Neural network prediction
- Noisy observations
- Physics loss training locations



equation Vlasov–Poisson solving for PINN

Networks(PINNs) Neural Informed Physics ۲

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Physics-informed neural networks for solving forward and inverse Vlasov–Poisson equation via fully kinetic simulation

Baiyi Zhang, Guobiao Cai, Huiyan Weng, Weizong Wang, Lihui Liu and Bijiao He

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[Machine Learning: Science and Technology, Volume 4, Number 4](#)

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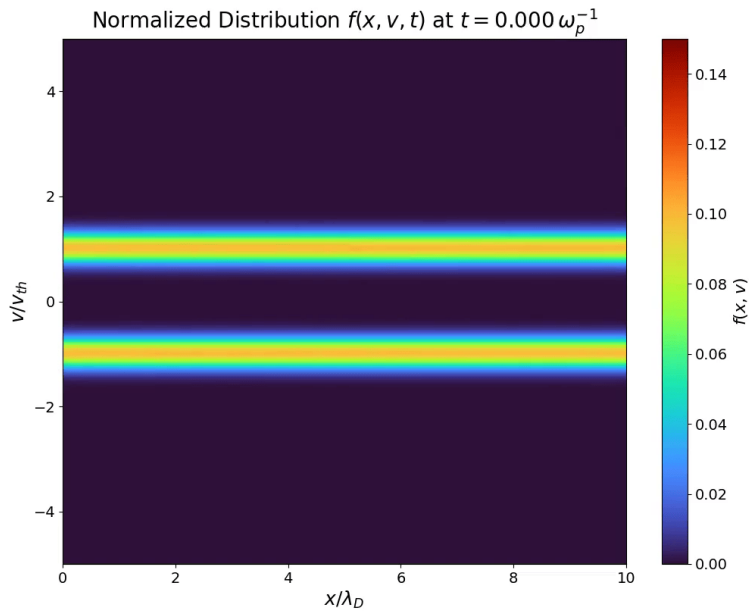
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Conclusion ►



instability stream two

Instability Stream Two ۳





equation Vlasov-Poisson

Instability Stream Two ۳

equation: Vlasov-Poisson

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla f + \frac{q}{m} (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \frac{\partial f}{\partial \mathbf{v}} = 0$$
$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon}$$

Density:



equation Vlasov-Poisson

Instability Stream Two ۳

equation: Vlasov-Poisson

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla f + \frac{q}{m} (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \frac{\partial f}{\partial \mathbf{v}} = 0$$
$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon}$$

Density:

$$\rho = e(n_e - Z_i n_i)$$
$$n_e = \int_{-\infty}^{+\infty} f(t, x, v) dv$$

equation: Vlasov-Poisson 1D



equation Vlasov-Poisson

Instability Stream Two ۳

equation: Vlasov-Poisson

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla f + \frac{q}{m} (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \frac{\partial f}{\partial \mathbf{v}} = 0$$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon}$$

Density:

$$\rho = e(n_e - Z_i n_i)$$

$$n_e = \int_{-\infty}^{+\infty} f(t, x, v) dv$$

equation: Vlasov-Poisson 1D

$$\frac{\partial f}{\partial t} + v \frac{\partial f}{\partial x} + \frac{eE}{m_e} \frac{\partial f}{\partial v} = 0$$

$$\frac{\partial E}{\partial x} = \frac{e(n_e - n_i)}{\varepsilon}$$



equation Vlasov–Poisson

Instability Stream Two ۳

$$x = \frac{x^*}{\lambda_D}, \quad v = \frac{v^*}{v_{th}}, \quad t = \frac{t^*}{\omega_p^{-1}}, \quad m = \frac{m^*}{m_e}, \quad q = \frac{q^*}{e}, \quad E = E^* \frac{e\lambda_D}{kT_e}$$

Normalization:



equation Vlasov-Poisson

Instability Stream Two ۳

Normalization:

$$x = \frac{x^*}{\lambda_D}, \quad v = \frac{v^*}{v_{th}}, \quad t = \frac{t^*}{\omega_p^{-1}}, \quad m = \frac{m^*}{m_e}, \quad q = \frac{q^*}{e}, \quad E = E^* \frac{e\lambda_D}{kT_e}$$

equation: Vlasov-Poisson 1D Normalized

$$\frac{\partial f}{\partial t} + v \frac{\partial f}{\partial x} - E \frac{\partial f}{\partial v} = 0$$

$$\frac{\partial E}{\partial x} = \int_{-\infty}^{+\infty} f(t, x, v) dv - 1$$



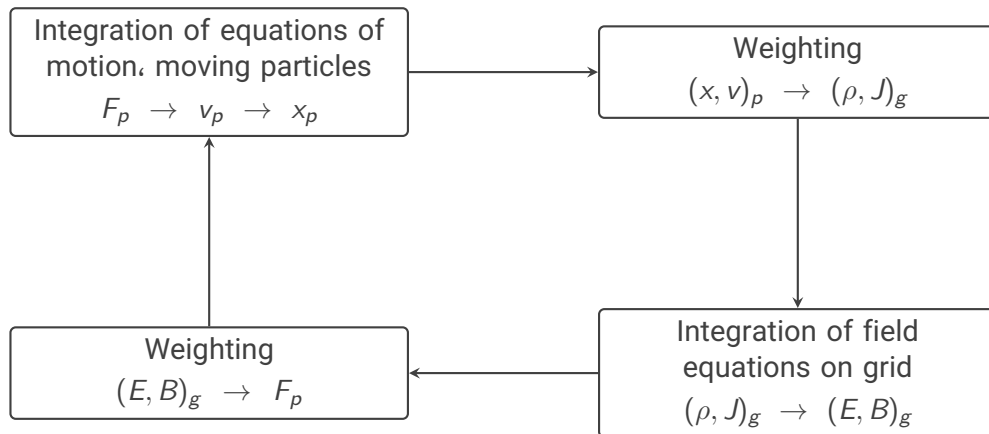
simulation PIC

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simulation PIC

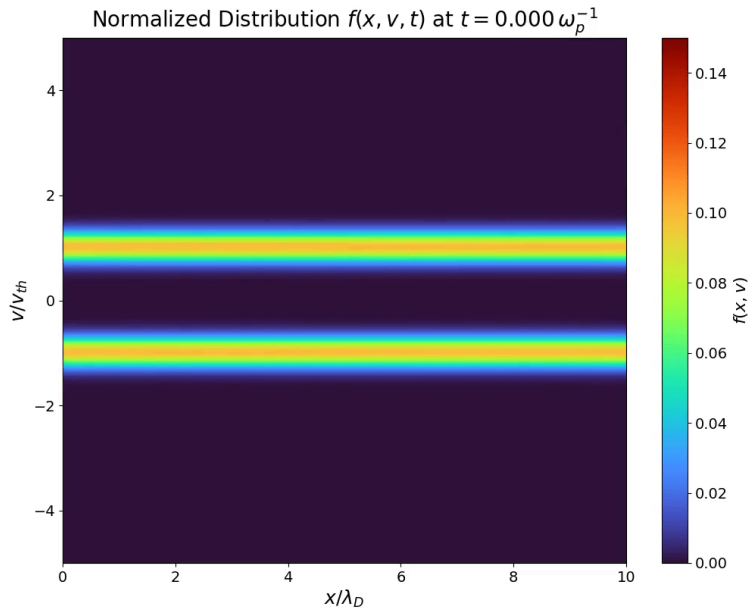
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instability stream two

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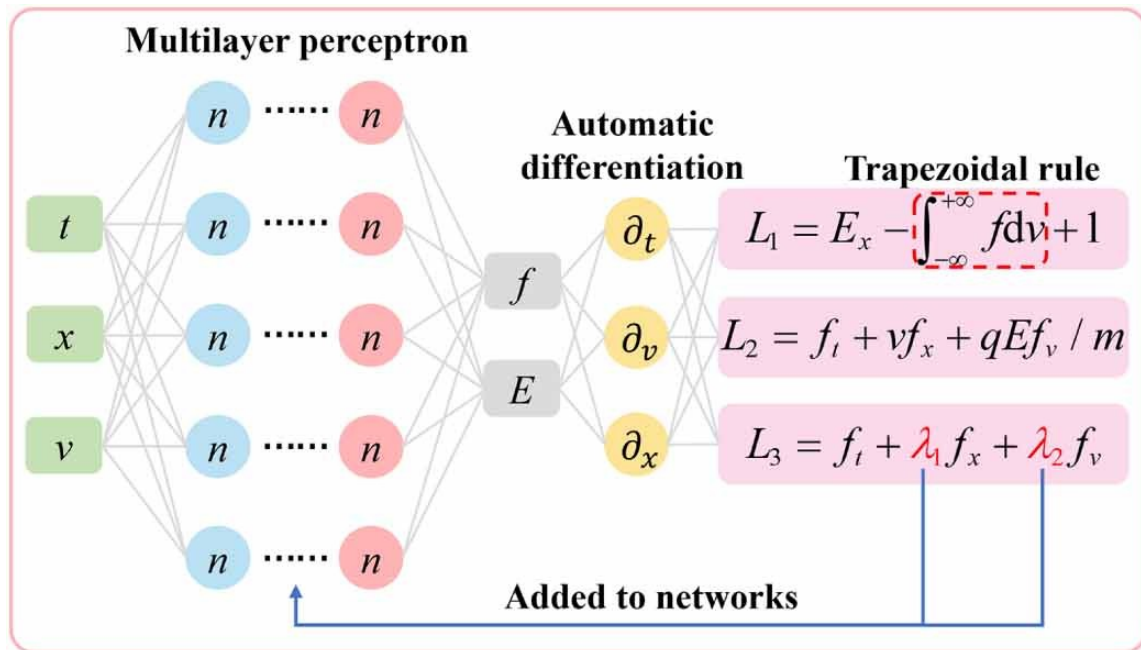
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PINN-Vlasov ۴





PINN-Vlasov Forward

PINN-Vlasov ۴

$$L = L_{\text{eq}} + L_{\text{IC}} + L_{\text{BC}}$$



PINN-Vlasov Forward

PINN-Vlasov ۴

$$L = L_{\text{eq}} + L_{\text{IC}} + L_{\text{BC}}$$

$$L_{\text{eq}} = \frac{1}{N_{\text{eq}}} \sum_{n=1}^{N_{\text{eq}}} \left[\left(\frac{\partial f}{\partial t} + v \frac{\partial f}{\partial x} + \frac{qE}{m} \frac{\partial f}{\partial v} \right)^2 + \left(\frac{\partial E}{\partial x} - \int_{-\infty}^{+\infty} f dv + 1 \right)^2 \right]_n$$

$$L_{\text{IC}} = \frac{1}{N_{\text{IC}}} \sum_{n=1}^{N_{\text{IC}}} \left[\left(f_{\text{IC}}^r - f_{\text{IC}}^t \right)^2 \right]_n$$

$$L_{\text{BC}} = \frac{1}{N_{\text{BC}}} \sum_{n=1}^{N_{\text{BC}}} \left[\left(f_{\text{BC}}^r - f_{\text{BC}}^t \right)^2 \right]_n$$



PINN-Vlasov Inverse

PINN-Vlasov ۴

$$L = L'_{\text{eq}} + L_{\text{f}}$$



PINN-Vlasov Inverse

PINN-Vlasov ۴

$$L = L'_{\text{eq}} + L_{\text{f}}$$

$$L'_{\text{eq}} = \frac{1}{N_{\text{eq}}} \sum_{n=1}^{N'_{\text{eq}}} \left[\left(\frac{\partial f}{\partial t} + \lambda_1 v \frac{\partial f}{\partial x} + \lambda_2 E \frac{\partial f}{\partial v} \right)^2 + \left(\frac{\partial E}{\partial x} - \int_{-\infty}^{+\infty} f dv + 1 \right)^2 \right]_n$$

$$L_{\text{f}} = \frac{1}{N_{\text{f}}} \sum_{n=1}^{N_{\text{f}}} \left[\left(f^{\text{r}} - f^{\text{t}} \right)^2 \right]_n$$



PINN-Vlasov Inverse

PINN-Vlasov ۴

$$L = L'_{\text{eq}} + L_{\text{f}}$$

$$L'_{\text{eq}} = \frac{1}{N_{\text{eq}}} \sum_{n=1}^{N_{\text{eq}}} \left[\left(\frac{\partial f}{\partial t} + \lambda_1 v \frac{\partial f}{\partial x} + \lambda_2 E \frac{\partial f}{\partial v} \right)^2 + \left(\frac{\partial E}{\partial x} - \int_{-\infty}^{+\infty} f dv + 1 \right)^2 \right]_n$$

$$L_{\text{f}} = \frac{1}{N_{\text{f}}} \sum_{n=1}^{N_{\text{f}}} \left[\left(f^{\text{r}} - f^{\text{t}} \right)^2 \right]_n$$

$$\lambda_1 = 1.0$$

$$\lambda_2 = \frac{q}{m} = \frac{-1.0}{1.0} = -1.0$$



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Challenges

Conclusion Δ

field electric the paper, the In **problem forward in model the to provided field Electric** $\blacktriangleleft$
 the limit may this training. forward during PINN the to supplied directly was $E(x, t)$
 physics. the from self-consistently E infer to ability model's

requires $\frac{\partial E}{\partial x} = \int f dv - 1$ term Poisson The **equation Poisson the in Integration** $\blacktriangleleft$
 affects rule) trapezoidal (e.g., method integration of choice The integration. numerical
 accuracy. and stability

term loss each of importance relative The **components loss of weighting Unknown** $\blacktriangleleft$
 bias or convergence hinder can weighting defined-improper clearly not is (L_{eq}, L_{IC}, L_{BC})
 network. the

training how on depends strongly performance The **strategy resampling and Sampling** $\blacktriangleleft$
 intervals resampling and frequency sampling Optimal (t, x, v) in chosen are points
 questions. research open remain



Challenges

Conclusion Δ

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 network. the

training how on depends strongly performance The **strategy resampling and Sampling** $\blacktriangleleft$
 intervals resampling and frequency sampling Optimal (t, x, v) in chosen are points
 questions. research open remain

Observation

frameworks. PINN-Vlasov for challenge main the still is efficiency training and stability. numerical accuracy. physical Balancing



Conclusion

Conclusion ◀

and learning deep between gap the bridge **(PINNs) Networks Neural Physics-Informed** ◀
modeling. physical
consistency. physical ensure they function. loss the into **PDEs governing** embedding By ◀
high with **problems inverse and forward** both solving for capability Demonstrated ◀
interpretability.



Work Future

Conclusion ♠

PDEs. complex for training faster — **efficiency computational Improving** ◀
points. collocation of selection dynamic — **sampling Adaptive** ◀
dimensions higher collisions. fields. magnetic include — **extension Multi-physics** ◀
(2D/3D).



Work Future

Conclusion ♡

PDEs. complex for training faster — **efficiency computational Improving** ◀
points. collocation of selection dynamic — **sampling Adaptive** ◀
dimensions higher collisions. fields. magnetic include — **extension Multi-physics** ◀
(2D/3D).

AI. Scientific Efficiency + Data + Physics = PINNs Future



Q&A

listening! for you Thank